Mock Paper A(2)

Started on	Thursday, 21 June 2018, 8:29 AM
State	Finished
Completed on	Thursday, 21 June 2018, 8:29 AM
Time taken	12 secs
Grade	**0.00** out of 100.00
Feedback	You are on the right track. Could you spend some more time on your revision?

Question 1

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 019
A hypertensive patient develops a small focal lesion of one inferior olivary nucleus. Which of the following clinical signs can be expected?

Select one:

- Right Left Disorientation
- Visual Inattention
- Anosognosia
- Oculomotor Apraxia
- Finger Nose Ataxia

Your answer is incorrect.

Inferior Olivary Nucleus has a role in motor coordination by providing CLIMBING FIBERS to the cerebellum.Lesion results in motor incoordination of the contralateral arm and leg (unlike other cerebellar lesions which cause ipsilateral dysfunction)

The correct answer is: Finger Nose Ataxia

Question 2

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 020
The Wisconsin Card SortingTest (WCST) assess which of the following functions?

Select one:

- category fluency
- personality traits
- visuomotor performance
- flexibility in problem solving
- Attention and concentration.

Your answer is incorrect.

The Wisconsin Card Sorting Test (WCST) is a neuropsychological test of 'set-shifting', i.e. the ability to display flexibility in the face of changing schedules of reinforcement

The correct answer is: flexibility in problem solving

Question 3

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 022
A patient was asked to name as many animals as possible in one minute. This test is useful to detect deficits involving which of the following structures?

Select one:

- Occipital lobe
- Cerebellum
- Temporal lobe
- Parietal lobe
- Frontal lobe

Your answer is incorrect.

'Category fluency test' is a frontal lobe function test for lexical fluency. The task involves listing names of animals or listing the items to buy in a supermarket. It tests not only the speed and accuracy but also the ability to shift from a narrow set to the next.

The correct answer is: Frontal lobe

Question 4

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 023
A 65-year-old lady has been diagnosed with normal pressure hydrocephalus. Which of the following symptoms is more likely to improve following shunting?

Select one:

- Executive functions
- Visuospatial deficits
- Cognitive impairment
- Urinary incontinence
- Gait impairment

Your answer is incorrect.

Gait disturbance is more likely to improve following ventriculo-peritoneal shunting in NPH.

The correct answer is: Gait impairment

Question 5

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 024
Raven's progressive matrices is used to assess

Select one:

- Frontal lobe functions
- Premorbid intelligence
- Visuospatial ability
- General intellectual ability
- Attentional bias

Your answer is incorrect.

Raven's Progressive Matrices (often referred to simply as Raven's Matrices) are multiple-choice intelligence tests of abstract reasoning that provide culturally unbiased IQ estimates. Rey Osterrieth test comprises of copying a complex figure initially and then reproducing it from immediate and delayed recall that tests both

visual memory and constructional ability

The correct answer is: General intellectual ability

Question 6

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 025
In a psychological test, the clinican reads an incomplete sentence 'The captain wanted to stay with the sinking?' and asks the patient to complete the sentence. Which of the following test is being carried out?

Select one:

- Wisconsin Test
- Hayling's Test
- Goldstein's Test
- Brixton Test
- Stroop test

Your answer is incorrect.

The Hayling Sentence Completion Test takes about five minutes to administer and yields three different measures of executive functioning. It consists of two sets of 15 sentences each having the last word missing. In the first section, the examiner reads each sentence aloud, and the participant has to simply complete the sentences (response initiation speed).The Brixton Spatial Anticipation Test measures the ability to detect rules in sequences of stimuli. It is perceptually simple and does not require a verbal response. It is thus appropriate for people who are suffering from a wide range of associated deficits such as those involving speech production or reading.

The correct answer is: Hayling's Test

Question 7

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 026
Lesions of the dominant parietal lobe may produce Gerstmann's syndrome. Which of the following is not a feature of this syndrome?

Select one:

- Hemi-neglect
- Acalculia
- Left-right confusion
- Finger agnosia
- Agraphia

Your answer is incorrect.

Hemi neglect is not a part of the tetrad described in Gerstmann syndrome.

The correct answer is: Hemi-neglect

Question 8

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 027
Which of the following movement disorders is characterized by involuntary flinging movements of the extremities?

Select one:

- Chorea
- Hemiballismus
- Writer's cramp

- Dysdiodochokinesia
- Athetosis

Your answer is incorrect.

Hemiballismus is usually a unilateral, nonrhythmic, rapid, nonsuppressible, violent, flinging movement of the proximal arm. Hemiballismus is caused by a lesion, usually an infarct, around the contralateral subthalamic nucleus. Although disabling, hemiballismus is usually self-limited, lasting 6 to 8 wk. Treatment with antipsychotics is often effective.

The correct answer is: Hemiballismus

Question 9

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 028
A 45-year-old gentleman suffers from anterograde memory loss with preserved insight. What is the most likely diagnosis?

Select one:

- Lesions involving orbitofrontal cortex
- Psychogenic amnesia
- Picks disease
- Diencephalic amnesia
- Hippocampal amnesia

Your answer is incorrect.

Anterograde amnesia is seen most frequently in patients who have sustained injuries to the medial temporal lobes or the diencephalon (Graff-Radford, Tranel, Von Hoesen, and Brandt, 1990). The medial temporal lobe contains the amygdala and the hippocampal formation, and animal experiments in primates have shown that damage to these two areas causes more severe anterograde amnesia. These two areas also send fibers to the diencephalon, and it is unclear if damage to the diencephalon affects the hippocampus and amygdala also.

The correct answer is: Hippocampal amnesia

Question 10

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 029
Many medical conditions are associated with autistic behavioural phenotype. Which one of the following is NOT associated with autism?

Select one:

- Fragile X Syndrome
- Tuberous sclerosis
- Huntington's disease
- Neurofibromatosis
- Down Syndrome

Your answer is incorrect.

Fragile-X, tuberous sclerosis, neurofibromatosis, hypomelanosis of Ito, Moebius syndrome, Rett syndrome, and Down syndrome are some of the medical conditions associated with autism.

The correct answer is: Huntington's disease

Question 11

Not answered

Marked out of 1.00

Flag question

MoK Applied Neuroscience 030
The negative symptoms of schizophrenia are related to which of the following neurobiological phenomena?

Select one:

- Hyperactivation of corpus striatum
- Hypoactivation of mesolimbic pathway
- Hyperactivation of dorsolateral prefrontal cortex
- Hypoactivation of dorsolateral prefrontal cortex
- Hypoactivation of corpus striatum

Your answer is incorrect.

It has been suggested that positive symptoms could be associated with excessive DA transmission in subcortical regions, whereas negative symptoms could be related to a concomitant deficit in cortical DA transmission. Earlier fMRI investigations, without pharmacologic challenge, have found some evidence for the concept of frontal hypoactivity, by showing reduced activation during prefrontal cognitive tasks (working memory, attention, and executive functions) in the ventrolateral prefrontal cortex (VLPFC), dorsolateral prefrontal cortex (DLPFC), and anterior cingulate in nonmedicated or medication-nave patients, relative to healthy controls.

The correct answer is: Hypoactivation of dorsolateral prefrontal cortex

Question 12

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 025
You are seeing a patient in the gender identity clinic. The patient would like to obtain temporary membership of the opposite sex. The most probable diagnosis is;

Select one:

- Body dysmorphic disorder
- Transexualism
- Dual role transvestism
- Frotteurism
- Fetishistic transvestism

Your answer is incorrect.

Dual role transvestism is a milder form of gender dysphoria. The individual wears clothes of the opposite sex in order to experience temporary membership in the opposite sex. The individual experiences a sense of appropriateness by wearing clothes of the other gender. There is no sexual motivation for the cross-dressing. The individual has no desire for a permanent change to the opposite sex.

The correct answer is: Dual role transvestism

Question 13

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 026
The presence of apolipoprotein-e4 allele increases the chance of developing Alzheimer's disease by around

Select one:

- Twenty fold
- Two fold
- Fifty fold
- Ten fold
- Five fold

Your answer is incorrect.

In 1993, a polymorphism in the apolipoprotein E4 allele (apoE4) was discovered to be a major risk factor for late- onset Alzheimer disease. This gene promotes the deposition of beta-amyloid protein in the brain, contributing to the formation of senile plaques. People who have inherited a single E4 allele of apolipoprotein (APOE) experience up to a three to fivefold increase in risk of developing Alzheimer disease.

The correct answer is: Five fold

Question 14

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 027
Culturing cells in conditions of thymidine deprivation can aid in the diagnosis of which of the following disorders?

Select one:

- Fragile X syndrome
- Asperger's disease
- Cri-du-chat syndrome
- Autism
- Down's syndrome

Your answer is incorrect.

Thymidine deprivation can make the fragile sites of the chromosome obvious for detection

The correct answer is: Fragile X syndrome

Question 15

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 028
Social anxiety and gaze avoidance are features of which of the following trinucleotide expansion syndrome?

Select one:

- Fredreich's Ataxia
- Leber's Optic Neuropathy
- Myotonic dystrophy
- Fragile X syndrome
- Huntington's disease

Your answer is incorrect.

Eye-gaze avoidance is a striking phenotypic feature in fragile X syndrome.

The correct answer is: Fragile X syndrome

Question 16

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 029
One of the following disorders do not involve genetic deletion. Choose the correct answer.

Select one:

- William syndrome
- Angelman syndrome
- Cri-du-chat syndrome
- DiGeorge syndrome
- Frederich's ataxia

Your answer is incorrect.

Friedreich's ataxia is caused by a defect in a gene called Frataxin (FXN), which is located on chromosome 9. Changes in this gene cause the body to produce too much of a part of DNA called trinucleotide repeat (GAA). Normally, the body contains about 8 to 30 copies of GAA. Those with Individuals with Friedreich's ataxia have as many as 1,000 copies. The more copies of GAA a patient has, the earlier in life the disease starts and the faster it gets worse. Friedreich's ataxia is an autosomal recessive genetic disorder [Friedreich's ataxia: MedlinePlus Medical Encyclopedia, http://www.nlm.nih.gov/medlineplus/ency/article/001411.htm].

The correct answer is: Frederich's ataxia

Question 17

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 030
Which one of the following is a primary proponent of endophenotypes concept?

Select one:

- Gottesmann
- Freud
- Kraeplin
- Fromm-Reichmann
- Kendler

Your answer is incorrect.

Gottesman and Shields re-introduced the endophenotypes concept.

The correct answer is: Gottesmann

Question 18

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 031
The process of meiosis produces

Select one:

- Diploid somatic cells
- Haploid germ cells
- Tetraploid cells
- Triploid cells
- Aneuploid cells

Your answer is incorrect.

The outcome of meiosis is four (genetically unique) haploid cells, compared with the two (genetically identical) diploid cells produced from mitosis. Meiosis is thus a reduction division where one diploid cell with two copies of each chromosome yields four haploid cells containing a single copy of each chromosome.

The correct answer is: Haploid germ cells

Question 19

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 032
Tay-Sachs disease is caused by deficiency of

Select one:

- Neuropeptide Y
- Glucoronidase

- Monoamine oxidase
- Hexosaminidase A
- Hypoxanthine phosphoribosyl transferase

Your answer is incorrect.

Tay-Sachs disease is caused by a deficiency of the enzyme Hexosaminidase. It is inherited as an autosomal recessive disorder. Clinical features are apparent by around six months of age and usually results in death by the age of four.

The correct answer is: Hexosaminidase A

Question 20

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 033
Which of the following is a characteristic feature of autosomal recessive transmission?

Select one:

- Pure maternal transmission
- Everyone with mutant gene is diseased
- High male to male transmission
- Phenotype not expressed in females
- High risk with consanguinity

Your answer is incorrect.

As carriers are possible in AR transmission, 'hidden relatives' may give birth to unwell children.

The correct answer is: High risk with consanguinity

Question 21

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 034
The anticipation in paternal transmission is most likely to show up in which of the following conditions?

Select one:

- Noonan syndrome
- Hurler syndrome
- Lesch-nyhan syndrome
- Huntington's disease
- Tuberous sclerosis

Your answer is incorrect.

Anticipation refers to an inheritance pattern in which individuals in the most recent generations of a pedigree develop a disease at an earlier age or with greater severity than does those in previous generations. This is mostly due to the gradual expansion of trinucleotide repeat polymorphisms (this instability is called dynamic mutation). The phenomenon of anticipation is seen in Huntington's disease. The inheritance of HD from the father is associated with a greater expansion of the repeats and an earlier age of onset. In approximately one-third of cases in which the father passes on the gene to his offspring, there is an expansion resulting in juvenile-onset HD. Characteristic protein deposits form nuclear inclusions in neurons of HD patients.

The correct answer is: Huntington's disease

Question 22

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 036
If working memory deficit is inherited as a multifactorial trait then its recurrence risk will be

Select one:

- same in all races
- not dependent on number of affected family members
- influenced by the severity of deficit
- same in all families
- same in both sexes

Your answer is incorrect.

Involvement of more family members and higher severity of expression of a multifactorial trait may mean higher polygenic contribution to the inheritance. Ethnic and gender differences are well noted in multifactorial inheritances.

The correct answer is: influenced by the severity of deficit

Question 23

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 037
Which of the following is most likely with regard to Fragile X syndrome?

Select one:

- Locus q25 is deleted
- Microcephaly is often seen
- Clinical features are apparent only during adolescence
- CTG repeats are notable
- Intelligence is correlated to the load of chromosomal damage

Your answer is incorrect.

CGG repeats at q27 of X chromosome is associated with macrocephaly and mental retardation during childhood.

The correct answer is: Intelligence is correlated to the load of chromosomal damage

Question 24

Not answered

Marked out of 1.00

Flag question

MoK Molecular Genetics 038
Which of the following refers to the stage between two successive cell divisions wherein the cell replicates its genetic material?

Select one:

- Metaphase
- Interphase
- Prophase
- Anaphase
- Telophase

Your answer is incorrect.

Interphase is the 'holding' stage

The correct answer is: Interphase

Question 25

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 007
When experiencing disgust, which of the following parts of brain show consistent changes in functional imaging?

Select one:

- Anterior insular region
- Supplementary motor area
- Dorso medial thalamus
- Posterior hippocampus
- Anterior cingulate cortex

Your answer is incorrect.

The involvement of insula in disgust recognition has been reported in behavioural and functional imaging studies. Both strong and mild expressions of disgust activated the same sites in the anterior insula.

The correct answer is: Anterior insular region

Question 26

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 013
Which of the following is seen in anterior cerebral artery occlusion?

Select one:

- Contralateral lower limb weakness
- Sensory aphasia
- Unilateral sensory loss
- Ipsilateral facial palsy
- Bulbar palsy

Your answer is incorrect.

Occlusion of the anterior cerebral artery may result in weakness and sensory loss of the contralateral foot and leg; gait apraxia and urinary incontinence (with bilateral damage in the acute phase). This sometimes resembles the triad of Normal Pressure Hydrocephalus where identical anatomical structures are implicated albeit presenting with an insidious or subacute onset. In left-sided lesions, some cases present with transcortical motor aphasia.

The correct answer is: Contralateral lower limb weakness

Question 27

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 014
A 56-year-old gentleman suffers from ataxia and nystagmus. His CT scan shows ischemia in cerebellum. Which part of cerebellum is most likely to be affected?

Select one:

- Vermis
- Anterior lobe
- Flocculo-nodular lobe
- Posterior lobe
- Lateral lobe

Your answer is incorrect.

Damage to the flocculonodular lobe (the vestibular part) may show up as a loss of equilibrium and, in particular, an altered walking gait, with a wide stance that indicates difficulty in balancing. Damage to the lateral zone, or the cerebrocerebellum, results in problems with skilled voluntary and planned movements. This can cause manifestations like hypotonia, dysarthria and dysdiadochokinesia and tremors.

The correct answer is: Flocculo-nodular lobe

Question 28

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 015
Which area of the brain is implicated with recognition of faces?

Select one:

- Amygdala
- Fusiform gyrus
- Nucleus accumbens
- Prefrontal cortex
- Left inferior frontal area

Your answer is incorrect.

Face perception is an ability that involves many areas of the brain. Brain imaging studies have typically shown a great deal of activity in the area of the temporal lobe known as fusiform gyrus, an area also known to cause prosopagnosia when damaged bilaterally.

The correct answer is: Fusiform gyrus

Question 29

Not answered

Marked out of 1.00

Flag question

MoK Neuroanatomy 016
Communication between the frontal lobes of each side of the brain is mediated through which of the following structures?

Select one:

- Fornix
- Genu of corpus callosum
- Parahippocampal gyrus
- Splenium of corpus callosum
- Arcuate fasciculus

Your answer is incorrect.

The corpus callosum forms the major connection by which the hemispheres communicate with one another. The genu of the corpus callosum contains fibers that pass from the frontal lobe of one side to that of the other.

The correct answer is: Genu of corpus callosum

Question 30

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 011
Which of the following enzymes is responsible for the degradation of noradrenaline?

Select one:

- acetylcholinesterase
- Choline Acetyl Transferase
- Catechol-O-Methyl transferase

- Tryptophan hydroxylase
- Tyrosine hydroxylase

Your answer is incorrect.

Catechol-O-methyltransferase and monoamine oxidase are critical for the metabolic degradation of catecholamines

The correct answer is: Catechol-O-Methyl transferase

Question 31

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 012
Which of the following substance is an active ingredient in the Ethiopian plant Catha edulis, whose leaves on chewing provide amphetamine-like stimulation?

Select one:

- Coumarin
- Cathine
- Ketamine
- Ephedrine
- Cathinone

Your answer is incorrect.

The stimulant effect of the Khat plant was originally attributed to cathine, but this was disputed by reports showing the plant extracts from fresh leaves contained another substance more active than cathine - called cathinone. Cathinone is not very stable and breaks down to produce cathine and norephedrine. This explains why fresh leaves are more stimulating, and the habit of chewing Khat is more or less confined to the regions where it is cultivated.

The correct answer is: Cathinone

Question 32

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 013
Hyperpolarization of the neuron is influenced by which of the following?

Select one:

- chloride and sodium
- calcium and chloride
- sodium and calcium
- potassium and sodium
- chloride and potassium

Your answer is incorrect.

In CNS neurons, inhibitory transmitters open chloride channels. In addition, second messengers that mediate inhibition work by operating on potassium channels.

The correct answer is: chloride and potassium

Question 33

Not answered

Marked out of 1.00

MoK Neurochemistry 014
The receptors which have both pre and postsynaptic presence are

Select one:

Flag question

- 5-HT2C receptors
- D2 receptors
- D1 receptors
- D5 receptors
- 5-HT1A receptors

Your answer is incorrect.

D2-like receptors (D-2, D-3 and D-4) have both pre-synaptic and post-synaptic inhibitory effect. D1 like receptors (D1 and D5) have only postsynaptic inhibition effect.

The correct answer is: D2 receptors

Question 34

Not answered

Marked out of 1.00

Flag question

MoK Neurochemistry 015
Which of the following is the most powerful endogenous opioid peptide?

Select one:

- ß-Endorphin
- Morphine
- Dynorphin
- Enkephalin (?-endorphin)
- Neoendorphin

Your answer is incorrect.

Dynorphin, a natural brain endorphin, is far more potent than morphine or any other known endorphin. With such high potency, dynorphin can bind with great specificity to pain receptors. Dynorphins, endorphins, and enkephalins arise from different biosynthetic pathways.

The correct answer is: Dynorphin

Question 35

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 007
All of the following intrauterine infections cause learning difficulties except

Select one:

- CMV
- Toxoplasmosis
- Treponema pallidum
- Rubella
- Epstein Barr Virus

Your answer is incorrect.

EBV (Infectious Mononucleosis) has not been implicated in learning disability syndromes. In pregnant women, measles (rubeola) unlike German measles (rubella) does not cause any congenital anomalies / learning disabilities.

The correct answer is: Epstein Barr Virus

Question 36

MoK Neuropathology 008

Not answered

Marked out of 1.00

Flag question

Which of the following is least likely to occur in schizophrenia?

Select one:

- Increase in neuron density
- Cytoarchitectural anomalies
- Reduced cell size affecting neurons
- Gliosis
- Reduced cell number affecting neurones

Your answer is incorrect.

In terms of histological changes, no evidence for astrogliosis in schizophrenia has been reported. Reduced cell numbers or cell size has been described especially affecting neurons in the hippocampus and dorsolateral prefrontal cortex. The Increase in neuronal density, which may relate to the observed decrease in neurone size has been reported. Subtle cytoarchitectural anomalies were described in the hippocampal formation and frontal cortex. Synaptic studies in the hippocampus and DLPFC in schizophrenia show decrements in presynaptic markers. (Ref: Harrison PJ. The neuropathology of schizophrenia. A critical review of the data and their interpretation. Brain 1999; 122: 593-624)

The correct answer is: Gliosis

Question 37

Not answered

Marked out of 1.00

Flag question

MoK Neuropathology 009
An intravenous heroin user presents with significant slowing of cognitive ability. He has fatuous affect and impaired concentration that has developed insidiously. Which of the following is a likely diagnosis?

Select one:

- Opioid intoxication
- Bacterial encephalitis
- Heroin withdrawal
- HIV dementia
- Depression

Your answer is incorrect.

HIV dementia can present with the above features. See Neuroscience section on HIV

The correct answer is: HIV dementia

Question 38

Not answered

Marked out of 1.00

Flag question

MoK Neurophysiology 005
The term used to describe EEG more than 13Hz is

Select one:

- Mu waves
- Beta waves
- Gamma waves
- Theta waves
- Alpha waves

Your answer is incorrect.

more than 13 = beta; 8 - 13 = alpha; 4-8 = theta; less than 4 = delta

The correct answer is: Beta waves

Question 39
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 008
The disorder associated with female pseudo-hermaphroditism is

Select one:
- Criduchat syndrome
- Cushing's syndrome
- Conn's syndrome
- Congenital adrenal hyperplasia
- Cotard's syndrome

Your answer is incorrect.

Therefore, female pseudohermaphroditism is a condition in which the gonadal sex is female, but the external genitalia are masculinized, caused by high androgen levels. This is most commonly caused by hyperplasia of the adrenal gland as a result of the genetic defects of steroidogenesis, called the virilizing congenital adrenal hyperplasias (CAHs)

The correct answer is: Congenital adrenal hyperplasia

Question 40
Not answered
Marked out of 1.00
Flag question

MoK Neurophysiology 009
A 58 year old man presents with postural hypotension, anorexia, nausea and weakness. He appears to be extremely apathetic, with significant anhedonia compounded by fatigue and depressed mood.Which of the following hormonal imbalance is most likely?

Select one:
- Vasopressin
- Insulin
- Thyroxine
- Corticosteroids
- Adrenaline

Your answer is incorrect.

Addison's syndrome may result from adrenal destruction by autoimmune disease, metastatic cancer, tuberculosis, or human immunodeficiency virus infection or secondary to suppression of ACTH secretion by chronic corticosteroid therapy. Symptoms include postural hypotension, anorexia, nausea and vomiting, and weakness. Psychiatric symptoms include apathy, anhedonia, fatigue, and depressed mood. Adrenal insufficiency can be mistaken for depressive disorder. Though a loss of appetite is seen in both, vomiting is almost never due to isolated depression. http://www.primarypsychiatry.com/aspx/articledetail.aspx?articleid=329

The correct answer is: Corticosteroids

Question 41
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 006
What is the incidence of fatal hepatotoxicity with valproate?

Select one:
- 1 in 250
- 1 in 100
- 1 in 100,000

- 1 in 1,000
- 1 in 10,000

Your answer is incorrect.

In those taking valproate alone, 1 in 100,000 patients. Risk factors for hepatotoxicity include young age (less than ten years old), sensitivity during the first few months of treatment, severe epilepsy, and use of multiple drugs.

The correct answer is: 1 in 100,000

Question 42

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 023
A 65-year-old man develops recurrent UTI since starting an antidepressant. He also gives a history suggestive of new onset prostatism. But prostate examination shows no enlargement. The most likely offending agent is

Select one:

- Nefazadone
- Reboxetine
- Trazodone
- Mirtazapine
- Amitryptyline

Your answer is incorrect.

Drugs with significant anticholinergic properties such as amitriptyline are notorious to cause signs of prostatism in elderly males. They may produce urinary retention and sometimes bladder distension.

The correct answer is: Amitryptyline

Question 43

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 024
Tom is an inpatient on a psychiatric ward. He has a history of psychotic depression and is on combination treatment with antipsychotics and antidepressants. He appears slightly flushed, with dry mouth, dry skin, feeling hot, and confused. He is exhibiting features of

Select one:

- SSRI discontinuation syndrome
- Neuroleptic malignant syndrome
- Idiopathic parkinsonism
- Anticholinergic syndrome
- Serotonin syndrome

Your answer is incorrect.

Anticholinergic syndrome results from the inhibition of cholinergic neurotransmission at muscarinic receptor sites. Psychotropics such as phenothiazines, tricyclic antidepressants and antiparkinsonian agents have anticholinergic properties. When they are administered in combination, the action of these drugs becomes additive resulting in anticholinergic syndrome. Features of a mild organic brain syndrome, which includes disorientation, difficulty in concentrating and impaired short-term memory are seen, especially noticeable at nighttime. Other physical symptoms include flushing, dry skin and mucosa, mydriasis with loss of accommodation and fever (Ref: Kaplan & Sadock Synopsis of Psychiatry 9th ed. Pg 1170-1172)

The correct answer is: Anticholinergic syndrome

Question 44

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 029
Patients on tranylcypromine should not take an excess of

Select one:

- Swede
- Banana
- Cheese
- Potatoes
- Salt

Your answer is incorrect.

Cheese can induce tyramine related reaction in those on MAOIs.

The correct answer is: Cheese

Question 45

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 030
Which of the following antipsychotics is associated with photosensitivity more than others listed?

Select one:

- Olanzapine
- Quetiapine
- Haloperidol
- Chlorpromazine
- Risperidone

Your answer is incorrect.

Allergic dermatitis and photosensitivity can occur with low-potency agents. Long-term chlorpromazine use can cause blue-gray discoloration of skin areas exposed to sunlight. This is reversible. Irreversible retinal pigmentation is associated with the use of high dose thioridazine (above 1000 mg a day).

The correct answer is: Chlorpromazine

Question 46

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 033
A 56-year-old gentleman suffers from a severe depressive episode. He has a history of bipolar affective disorder. He is on maintenance treatment with Lithium for five years. What is the most appropriate drug to treat his depression?

Select one:

- Amitriptyline
- Venlafaxine
- Phenelzine
- Citalopram
- Duloxetine

Your answer is incorrect.

The risk of inducing a switch to mania is less with SSRIs and more with other classes of antidepressants such as TCAs, MAOIs and SNRIs.

The correct answer is: Citalopram

Question 47
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 034
Which of the following drugs is used in the treatment of akathisia?

Select one:
- Procyclidine
- Atrophine
- Trihexyphenidyl
- Clonazepam
- Benperidol

Your answer is incorrect.

Treatments for akathisia include reducing the antipsychotic drug or change to an atypical drug. A reduction in symptoms of akathisia may be seen with propranolol, clonazepam, mirtazapine, trazodone, mianserin and cyproheptadine

The correct answer is: Clonazepam

Question 48
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 035
Which of the following drugs can cause severe rebound hypertension when stopped?

Select one:
- risperidone
- venlafaxine
- duloxetine
- paroxetine
- clonidine

Your answer is incorrect.

Clonidine is an Alpha-2 adrenergic receptor agonist. Agonist effects at these receptors, which are located presynaptically in adrenergic neurons, reduce the release of epinephrine from these neurons.

The correct answer is: clonidine

Question 49
Not answered
Marked out of 1.00
Flag question

MoK Adverse Effects 036
Tom is a 10-year-old boy with Tourettes' syndrome. He has had a medication for treating his condition, which has lead to significant weight gain. Choose the best alternative?

Select one:
- Risperidone
- Clonidine
- Haloperidol
- Pimozide
- Olanzapine

Your answer is incorrect.

Weight gain may be more pronounced in children than adults with antipsychotics, and this may limit their use. Clonidine has been shown in open studies to reduce the severity and frequency of tics and could bring a substantial reduction in tics. (Ref; Leckman JF et al.; Arch Gen Psychiatry;1991: 42: 324-328)

The correct answer is: Clonidine

Question 50

Not answered

Marked out of 1.00

Flag question

MoK Adverse Effects 037
Which of the following statement regarding lithium is NOT true?

Select one:

- Oral preparations may have different bioavailability
- Hypothyroidism is more common in middle-aged women talking lithium
- ECG changes can be seen even at therapeutic doses
- Coarse tremor is a common side effect
- Peripheral limb oedema is a recognised side effect of lithium use

Your answer is incorrect.

Lithium chloride causes a concentration dependent block of the peak sodium current in cardiac pacemakers, but even at therapeutic doses, abnormal ECG can be documented, often with no clinical implications. Fine hand tremor is a common side effect often seen at lower doses; coarse tremor indicates toxicity.

The correct answer is: Coarse tremor is a common side effect

Question 51

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 013
Co-agonists glycine and d-serine act on which of the following receptors?

Select one:

- AMPA
- Serotonergic
- Adenosine
- Muscarinic
- Kainate

Your answer is incorrect.

D-serine is an endogenous ligand for the glycine site of the N-methyl-D-aspartate receptor. Some recent studies suggest that there is a low affinity and stereo-selective site at the agonist-binding pocket of AMPA receptors as well for D-amino acids such as d-serine.

The correct answer is: AMPA

Question 52

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 014
The constituents of Khat have action that has greatest similarity to which of the following drugs?

Select one:

- Cocaine
- Phencyclidine
- Amphetamine
- LSD

- Cannabis

Your answer is incorrect.

The fresh leaves of Catha edulis from East Africa were used as a stimulant for amphetamine-like effects from the main ingredient cathinone. Cathinone has most of the CNS and peripheral actions of amphetamine. It is typically absorbed buccally after chewing the leaf and being an alkaloid, gets metabolized rapidly with low chances of toxicity.

The correct answer is: Amphetamine

Question 53

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 015
A 56-year-old woman on treatment with Clozapine has some residual symptoms and a BMI of 27. The ideal drug of choice for adjunctive treatment is

Select one:

- Olanzapine
- Citalopram
- Fluoxetine
- Quetiapine
- Aripiprazole

Your answer is incorrect.

The addition of Aripiprazole to Clozapine may be particularly effective in reversing metabolic effects (such as weight gain) and could be used as an augmenting agent to Clozapine.

The correct answer is: Aripiprazole

Question 54

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 016
The mechanism of action of Cyproterone acetate is

Select one:

- Blocks testosterone receptors
- LHRH antagonism
- Reduces Prolactin levels
- Agonist at testosterone receptors
- 17- alpha hydroxy catabolism

Your answer is incorrect.

The mechanism of action of Cyproterone acetate is to block testosterone receptors. The direct antiandrogenic effect of Cyproterone is to block the binding of dihydrotestosterone to the specific receptors in the prostatic carcinoma cell. In addition, cyproterone exerts a negative feed-back on the hypothalamo-pituitary axis, by inhibiting the secretion of LH resulting in diminished production of testicular testosterone (Retrieved from Archives of Product Info: http://www.drugs.com/mmx/cyproterone-acetate.html).

The correct answer is: Blocks testosterone receptors

Question 55

Not answered

Marked out of 1.00

MoK Pharmacodynamics 017
Which of the following drugs acts as a Partial Mu receptor agonist?

Select one:

Flag question

- Acamprosate
- Buprenorphine
- Disulfiram
- Methadone
- Buspirone

Your answer is incorrect.

Buprenorphine is a partial agonist at the mu-opioid receptor and has a reduced risk of fatal respiratory depression.

The correct answer is: Buprenorphine

Question 56

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 018
Which of the following atypical antidepressants have dopaminergic and noradrenergic actions?

Select one:

- Bupropion
- Mirtazapine
- Venlafaxine
- Reboxetine
- Mianserin

Your answer is incorrect.

Bupropion is an atypical antidepressants have dopaminergic and noradrenergic actions, and it has been advocated by NICE for smoking cessation

The correct answer is: Bupropion

Question 57

Not answered

Marked out of 1.00

Flag question

MoK Pharmacodynamics 019
The drug used in treatment of REM sleep behavioural disorder is

Select one:

- Citalopram
- Sildenafil
- Amitryptyline
- Olanzapine
- Clonazepam

Your answer is incorrect.

Treatment with clonazepam has shown to be effective in REM sleep behavioural disorders.

The correct answer is: Clonazepam

Question 58

Not answered

Marked out of 1.00

MoK Pharmacodynamics 020
Which of the following acts on presynaptic receptors as main mechanism of action?

Select one:

Flag question

- Phenelzine
- Propronalol
- Donepezil
- Diazepam
- Clonidine

Your answer is incorrect.

Clonidine is a sympatholytic medication, which is classified as a centrally acting a2 adrenergic agonist. It has specificity towards the presynaptic a2 receptors in the vasomotor center in the brainstem.

The correct answer is: Clonidine

Question 59

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 019
A patient has chronic schizophrenia and is on clozapine therapy. If you want to try augmentation, which one should be avoided?

Select one:

- Carbamazepine
- Sodium valproate
- Aripiprazole
- Fluoxetine
- Lithium

Your answer is incorrect.

The combination of carbamazepine and clozapine is generally contraindicated. First, carbamazepine and clozapine each produce hematologic side effects at higher frequencies than do most other available drugs. Another consideration is that carbamazepine lowers the level of clozapine, possibly more than it lowers levels of traditional neuroleptics. Am J Psychiatry 155:12-21, January 1998

The correct answer is: Carbamazepine

Question 60

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 020
Liz was started on a mood stabiliser when she was diagnosed with bipolar affective disorder type 2. Her psychiatrist says that the dose needs to be carefully titrated as the prescribed mood stabiliser induces its own metabolism. Which of the following has been prescribed to Liz?

Select one:

- Carbamazepine
- Semisodium valproate
- Lamotrigine
- Lithium
- Sodium valproate

Your answer is incorrect.

Carbamazepine has a tendency to induce its own metabolism leading to reduced plasma steady state levels.

The correct answer is: Carbamazepine

Question 61

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 021
Which of the following does not increase tricyclic antidepressant concentrations?

Select one:

- Cigarette smoking
- Haloperidol
- Quetiapine
- Methylphenidate
- Risperidone

Your answer is incorrect.

Antipsychotics, methylphenidate, acetazolamide, aspirin, cimetidine, thiazides, fluoxetine and sodium bicarbonate increase tricyclic antidepressant (TCA) concentrations. Cigarette smoking decreases their concentration through its action on the 1A2 enzyme

The correct answer is: Cigarette smoking

Question 62

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 022
The antidepressant with the longest mean half-life from the given list is

Select one:

- fluvoxamine
- venlafaxine
- Citalopram
- paroxetine
- trazodone

Your answer is incorrect.

Fluoxetine has the longest half-life of all the SSRIs, one to three days, because of a metabolite. Paroxetine, the SSRI with a long t1/2 compared to the others, has a half-life of only 21 hours

The correct answer is: Citalopram

Question 63

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 023
A patient who is prescribed disulfiram to aid alcohol abstinence, presents to Accident and Emergency room with facial flushing, hypotension, tachycardia, nausea, and vomiting. She denies any recent ingestion of alcohol. Which of the following could have caused her symptoms?

Select one:

- Walnuts
- Mature cheese
- Banana
- Chocolate
- Cough syrup

Your answer is incorrect.

Cough and cold medications bought over the counter often contain alcohol and can precipitate such Disulfiram reaction. Severe Antabuse reaction (also known as disulfiram reaction) is characterized by respiratory depression, seizures, cardiovascular collapse, and even death.

The correct answer is: Cough syrup

Question 64

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 024
A barbiturate-dependent patient shows little pharmacological response to administration of benzodiazepines. This is called

Select one:

- Tolerance
- Dependence
- Down regulation
- Sensitisation
- Cross tolerance

Your answer is incorrect.

Cross tolerance refers to decreased response to one medication due to exposure to a different medication. Here exposure to barbiturates has produced cross-tolerance to benzodiazepines.

The correct answer is: Cross tolerance

Question 65

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 025
The benzodiazepines with the longest half life in a young adult is;

Select one:

- Alprazolam
- Diazepam
- Clonazepam
- Nitrazepam
- Temazepam

Your answer is incorrect.

Diazepam is a benzodiazepine which has a very long elimination half-life (20 - 100 hours with an active metabolite 36 - 200 hours); as a result withdrawal symptoms are not very common with diazepam.

The correct answer is: Diazepam

Question 66

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 026
When administered orally, the benzodiazepine that produces the most rapid onset of action is

Select one:

- Alprazolam
- Lorazepam
- Diazepam.
- Chlordiazepoxide
- Nitrazepam

Your answer is incorrect.

Diazepam has the most rapid onset of action among orally administered benzodiazepines.

The correct answer is: Diazepam.

Question 67

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 027
You are treating an Afro-Caribbean man with Schizophrenia using clozapine. What is the biggest risk factor for developing reduced white cell count?

Select one:

- Male sex
- Ethnicity
- Dose related
- Duration of treatment
- Treatment setting

Your answer is incorrect.

There is a 77% increase in risk for Afro-Caribbean people to develop clozapine-induced neutropenia. Other risk factors include young age and having a low baseline white cell count.

The correct answer is: Ethnicity

Question 68

Not answered

Marked out of 1.00

Flag question

MoK Pharmacokinetics 028
A patient who needs a prescription for a new antidepressant is almost certain that he will miss one or two tablets during his treatment due to the nature of his job. What is the most suitable choice of medications?

Select one:

- escitalopram
- paroxetine
- citalopram
- venlafaxine
- Fluoxetine

Your answer is incorrect.

Due to its longer half-life, missing fluoxetine for a day or two will not usually cause acute withdrawal symptoms. This is a safe bet for the patient described in this question.

The correct answer is: Fluoxetine

Question 69

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI008
Choosing psychiatric medications
For each of the following descriptions choose the most appropriate antidepressant from the given list

A 24-year-old woman has had two episodes of irritable mood, increased energy, decreased need for sleep, and pressured speech. She was recently started on medication by her psychiatrist to control these symptoms. She presents to her GP now complaining of a significant worsening of her acne since starting this new medication. What drug was she started on?

Choose...

A 24-year-old man has multiple problems at his workplace due to 'extreme tiredness' during office hours. For several years, he has had severe daytime sleepiness, episodes of falling asleep without warning, and hearing his name being called out loud on many occasions when no one is around. Which of the following treatments may be of benefit to this patient? Choose...

Which of the following drugs carried a risk of thromboembolism especially during the first three months of treatment? Choose...

Your answer is incorrect.

Several cutaneous side effects are possible with lithium including acne, follicular and maculopapular eruptions. Alopecia has also been reported. With the exception of lithium, the other choices do not worsen acne.

Case 2: The description in the questions fits with a diagnosis of narcolepsy. Stimulants such as methylphenidate can reduce daytime sleepiness. Medications that reduce REM sleep, such as TCAs and SSRIs, are used if cataplexy is also present. Modafinil can reduce the number of sleep attacks and can improve psychomotor performance.

Clozapine's adverse effect of thromboembolism may be related to its effect on antiphospholipid antibodies; though rare, if it occurs, this is usually in the first three months of the treatment.

The correct answer is: A 24-year-old woman has had two episodes of irritable mood, increased energy, decreased need for sleep, and pressured speech. She was recently started on medication by her psychiatrist to control these symptoms. She presents to her GP now complaining of a significant worsening of her acne since starting this new medication. What drug was she started on? – Lithium, A 24-year-old man has multiple problems at his workplace due to 'extreme tiredness' during office hours. For several years, he has had severe daytime sleepiness, episodes of falling asleep without warning, and hearing his name being called out loud on many occasions when no one is around. Which of the following treatments may be of benefit to this patient? – Methylphenidate, Which of the following drugs carried a risk of thromboembolism especially during the first three months of treatment? – Clozapine

Question 70

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI016

Neuropsychological measurement

Identify the test, which could be used in each of the following situations

Which of the above tests use the following question: 'How many camels are there in England?' (Choose one) Choose...

Scores are given for pronouncing the words DRACHM, SUPERFLUOUS and PLACEBO correctly (Choose one) Choose...

The subject is asked to connect scattered numbers and alphabets in the sequence 1A2B3C4D5E6F..., using a pen and paper. (Choose one) Choose...

Your answer is incorrect.

In cognitive estimates test, factual questions that require abstract processing are asked. This is a test of frontal function. In NART, scores are given for correct pronunciation of complex phonetic words.- In trail making test A, the subject has to simply connect 1-2-3-4-5-6...etc as a sequence. In trail making test B, the subject has to connect the sequence 1A2B3C4D5E6F...etc.

The correct answer is: Which of the above tests use the following question: 'How many camels are there in England?' (Choose one) – Cognitive estimates test, Scores are given for pronouncing the words DRACHM, SUPERFLUOUS and PLACEBO correctly (Choose one) – National Adult Reading Test, The subject is asked to connect scattered numbers and alphabets in the sequence 1A2B3C4D5E6F..., using a pen and paper. (Choose one) – Trail making test B

Question 71

Not answered

Marked out of 3.00

MoK Pharmacology EMI010

Partial agonism

Identify the predominant receptor level of action for the following partial agonistic drugs.

Flag question

Buspirone Choose...

Clonazepam Choose...

Buprenorphine Choose...

Your answer is incorrect.

Partial agonist- Aripiprazole - D2; Buspirone 5HT1A; Clonazepam - BDZ receptor; Buprenorphine - opioid receptor mu.

Aripiprazole carries a low risk of extrapyramidal symptoms despite high levels of D2 occupancy (> 90%) at therapeutic doses; this is probably due to its partial-agonist activity at D2 receptors.

Buspirone acts as a partial agonist on serotonin 5-HT1A receptors - presynaptic agonism leads to inhibition of release of serotonin, with consequent antianxiety effects. Postsynaptic agonism leads to antidepressant activity.

Benzodiazepines act via a specific site called omega site in GABA-A complex. All are agonists except clonazepam, which is a partial agonist.

The correct answer is: Buspirone – 5 HT1A receptor, Clonazepam – Benzodiazepine receptor, Buprenorphine – Mu opioid receptor

Question 72

Not answered

Marked out of 3.00

Flag question

MoK Pharmacology EMI011

Chemical class of psychotropic drugs

Match each chemical class below with the most appropriate drug from the given list:

Thienobenzodiazepine Choose...

Dibenzodiazepine Choose...

Substituted benzamides Choose...

Your answer is incorrect.

Thienobenzodiazepine- Olanzapine;

Dibenzodiazepine- Clozapine; Substituted benzamides include Amisulpride, Sulpiride.

Dibenzothiazepine- Quetiapine.

Benzisoxazole derivatives include Risperidone;

Thioxanthenes include Flupenthixol and Zuclopenthixol.

The correct answer is: Thienobenzodiazepine – Olanzapine, Dibenzodiazepine – Clozapine, Substituted benzamides – Amisulpride

Question 73

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI011

Cerebral arteries & lesions

Using the clinical descriptions below, identify the artery most likely to be involved:

A 58-year-old gentleman presents with diplopia, ataxia, nystagmus on looking to the side of the lesion and Horner's syndrome Choose...

A 57-year-old gentleman who is a right handed individual presenting with right hemiparesis, dysphasia and sensory loss involving face & arms on the right side.

Choose...

A 67-year-old gentleman with presenting with right-sided weakness especially in the legs. The weakness is milder in the arm, and the face is spared.

Choose...

Your answer is incorrect.

The lateral medullary syndrome or posterior inferior cerebellar artery (PICA) thrombosis, or Wallenberg's syndrome, is a brain-stem infarction presenting as acute vertigo with cerebellar signs. It follows thromboembolism in the PICA or its branches, vertebral artery thromboembolism or dissection. On the side of the lesion, there could be facial numbness (5th nerve), diplopia (6th nerve), nystagmus, ataxia (cerebellar), Horner's syndrome, 9th& 10th nerve lesions. On the contralateral side, there could be spinothalamic sensory loss and mild if any, hemiparesis.

Middle cerebral artery lesions are characterized by hemiplegia, hemianesthesia. If the lesion involves dominant hemisphere, the patients often present with other symptoms such as dysphasia and dyspraxia. If the lesion involves non-dominant hemisphere, then contralateral neglect is highly likely.

Anterior cerebral artery lesions involve contralateral hemiplegia/hemiparesis with weakness involving more of the legs than the arms, and the face will usually be spared.

The correct answer is: A 58-year-old gentleman presents with diplopia, ataxia, nystagmus on looking to the side of the lesion and Horner's syndrome – Posterior inferior cerebellar artery, A 57-year-old gentleman who is a right handed individual presenting with right hemiparesis, dysphasia and sensory loss involving face & arms on the right side. – Left middle cerebral artery, A 67-year-old gentleman with presenting with right-sided weakness especially in the legs. The weakness is milder in the arm, and the face is spared. – Left anterior cerebral artery

Question 74

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI013

Familial risk rates

Choose the most appropriate risk in each of the following situations:

The risk of inheritance in a 30-year-old man whose mother and maternal grandfather developed Alzheimer's disease in their 50s

Choose...

A 25-year-old woman with a history of bipolar disorder. What is the chance that one of her children may develop a bipolar spectrum disorder?

Choose...

A 36-year-old woman with unipolar depression. What is the chance that her daughter inherits the same illness?

Choose...

Your answer is incorrect.

For the common late-onset form of Alzheimer's, the risk for those with no first degree relatives (parents or siblings) with a diagnosis of Alzheimer's disease is 7% or less (given normal life expectancy). The risk for those with one first-degree family member is about 15-20%. For second degree relatives (e.g., grandparents) the risk may be about 10%. For monozygotic twins, the risk is 80-85%. But in early-onset cases (<65) strong autosomal dominant tendency is noted increasing the risk to >50%.

Compared with the offspring of control parents, children of parents with bipolar disorder had an increased risk of having a bipolar spectrum disorder (Odds Ratio 13.4). A meta-analysis and more recent studies have reported rates of BP between 4% and 15% (B is the closest answer) in the offspring of parents with BP and between 0% and 2% in the offspring of healthy parents.(Lapalme et al. Can J Psychiatry. 1997;42(6):623-631). Children in families where both parents had bipolar disorders also were more likely than those in families containing one parent with bipolar disorder to develop the condition (Odds Ratio: 3.6) [Birmaher et al. Arch Gen Psychiatry. 2009;66(3):287-296. Children of parents with bipolar disorder have an increased risk of unipolar (11.4%) depression too. Children of parents with unipolar depression have a 5.9 to 18.4 % (median 9.1%: the closest answer is 10-19%) risk of unipolar depression and no increased risk of bipolar disorder.

The correct answer is: The risk of inheritance in a 30-year-old man whose mother and maternal grandfather developed Alzheimer's disease in their 50s – 50-59%, A 25-year-old woman with a history of bipolar disorder. What is the chance that one of her children may develop a bipolar spectrum disorder? – 5-10%, A 36-year-old woman with unipolar depression. What is the chance that her daughter inherits the same illness? – 10-19%

Question 75

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI015

Neuroimaging findings

For each of the condition given below choose ONE appropriate finding from the above list

Geriatric depression — Choose...

Variant CJD — Choose...

Auditory hallucinations in schizophrenia — Choose...

Your answer is incorrect.

In geriatric depression the common MRI findings include atrophy, ventricular enlargement, lesions in basal ganglia and deep white matter hyper intensities (WMH). WMH occur particularly in the deep subcortical white matter and to a lesser extent in the basal ganglia and periventricular white matter (Harrison, 2002). In major depression, WMH are linked to risk factors for vascular disease, and thus are often seen in the elderly subjects.

In variant CJD, MRI is the most useful supportive diagnostic test. A characteristic abnormality seen in the posterior thalamic region (pulvinar sign) is highly sensitive and specific for variant CJD. The pulvinar sign has been found in more than 90% of pathologically proven vCJD cases. FLAIR sequences of MRI are most likely to show the abnormality.

In patients with auditory hallucinations in Schizophrenia, primary auditory cortex activation is seen in the f-MRI scan (Dierks et al-1999). Neuroimaging studies have shown that auditory hallucinations in schizophrenic patients were associated with increased metabolic activity in the left primary auditory cortex and the right middle temporal gyrus.

Harrison, PJ (2002). The neuropathology of primary mood disorder. Brain, http://brain.oxfordjournals.org/content/125/7/1428

The correct answer is: Geriatric depression – Deep white matter hyperintensities (MRI), Variant CJD – Bilateral high pulvinar signals (MRI), Auditory hallucinations in schizophrenia – Primary auditory cortex activation (fMRI)

Question 76

Not answered

Marked out of 4.00

Flag question

MoK Neurosciences EMI017

Personality tests

For each of the following tests given below, choose their correct names from the list given:

This test is a way of measuring personality along the dimensions of extroversion-introversion and neuroticism-stability — Choose...

It is one of the objective personality tests that includes hypochondriasis as a personality factor — Choose...

A projective personality test with pictures of ambiguous persons and scenes to make up a story — Choose...

This test is useful to assess premorbid intelligence. — Choose...

Your answer is incorrect.

Eysenck's approach: Eysenck used second order analysis (orthogonal factor analysis) that identified small number of powerful independent factors relevant to measuring personality. This method yielded three-dimensional traits. These are neuroticism (vs. stability), psychoticism and extraversion (vs. introversion). Eysenck's personality questionnaire contains a lie scale. Eysenck maintained that no fundamental discontinuity exists between normal and abnormal personalities (dimensional).

Minnesota multiphasic personality inventory MMPI is a popular inventory for measuring personality. It has ten scales with clinical labels including hypochondriasis and hypomania. It is NOT a projective test.

Rorschach Inkblot Test and Murray's Thematic Apperception Test are projective tests; so are sentence completion and Draw a Person tests. Projective tests use ambiguous stimuli, and the patient is expected to generate narratives based on them; rating scales may or may not be used in conjunction to rate such measures.

NART - National adult reading test: Taps on previous word knowledge before becoming ill - hence used to estimate premorbid IQ.

The correct answer is: This test is a way of measuring personality along the dimensions of extroversion-introversion and neuroticism-stability – Eysenck Personality Inventory, It is one of the objective personality tests that includes hypochondriasis as a personality factor – Minnesota multiphasic personality inventory, A projective personality test with pictures of ambiguous persons and scenes to make up a story – Thematic apperception test, This test is useful to assess premorbid intelligence. – National adult reading test

Question 77

Not answered

Marked out of 3.00

Flag question

MoK Neurosciences EMI019

Derivatives of neural tube

For each of the following organs choose the neural tube vesicle of origin from the list provided:

Thalamus Choose...

Hippocampus Choose...

Cerebellum Choose...

Your answer is incorrect.

The telencephalon (cerebrum) together with the diencephalon constitute the forebrain. The diencephalon ("interbrain") is the region of the brain that includes the thalamus, metathalamus, hypothalamus, epithalamus, prethalamus or subthalamus and pretectum.The telencephalon or the cerebrum is composed of the following sub-regions:cerebral cortices, basal ganglia and the limbic system. Cerebellum is part of the metencephalon, which also includes the pons; the metencephalon is the upper part of the rhombencephalon or "hindbrain".

The correct answer is: Thalamus – Diencephalon, Hippocampus – Telencephalon, Cerebellum – Metencephalon

Question 78

Not answered

Marked out of 4.00

Flag question

MoK Neurosciences EMI027

Neuropathological features

You are working in a neuroscience department, and you have a special interest in dementias. For each of the definitions given below, identify the neuropathological terms used to describe them.

Weakly eosinophilic spherical cytoplasmic inclusions Choose...

This substance accelerates reuptake of dopamine in neurons that might be toxic Choose...

Abnormal swollen neuronal cells, with loss of Nissl's substance, oval in shape with peripherally displaced nucleus Choose...

These are masses of cytoskeletal elements, which are argentophilic, tau and ubiquitin reactive filamentous inclusions. Choose...

Your answer is incorrect.

Lewy bodies are weakly eosinophilic, spherical, cytoplasmic inclusions. In Parkinson's disease they are confined to substantia nigra; in Lewy body dementia, they are also present in many areas of the cerebrum including the temporal lobe, the cingulate gyrus and the frontal lobes. Lewy bodies in Parkinson's disease and DLB contain accumulations of alpha-synuclein.

Alpha-synuclein accelerates reuptake of dopamine in neurons, and this dopamine overload might be toxic. Lewy neurites are nerve cell processes that contain aggregates of alpha-synuclein. These abnormal structures can occur in both DLB/dementia of Parkinson's disease and idiopathic Parkinson's disease and are most numerous in the CA2/3 region of the hippocampus and the substantia nigra.

A disproportionate atrophy of the frontotemporal regions than other brain regions is seen in Pick's disease. These areas also show a reduction in large cortical nerve cells, abundant gliosis, and neuronal Pick's bodies, which are masses of cytoskeletal elements. Pick's bodies are argentophilic, tau and ubiquitin reactive filamentous inclusions. Abnormal swollen neuronal cells, with a loss of Nissl's substance, oval in shape with peripherally displaced nucleus are called Pick cells.

Ref: Love S., (2005) Neuropathological investigation of dementia: a guide for neurologists. Retrieved from http://jnnp.bmj.com/content/76/suppl_5/v8.full

The correct answer is: Weakly eosinophilic spherical cytoplasmic inclusions – Lewy bodies, This substance accelerates reuptake of dopamine in neurons that might be toxic – Alpha-synuclein, Abnormal swollen neuronal cells, with loss of Nissl's substance, oval in shape with peripherally displaced nucleus – Pick cells, These are masses of cytoskeletal elements, which are argentophilic, tau and ubiquitin reactive filamentous inclusions. – Pick bodies

Finish review